Hypospadias

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Definition

- ☑ Hypospadias can be defined as hypoplasia of the tissues forming the ventral aspect of the penis beyond the division of the corpus spongiosum.
- ☑ In simple terms hypospadias is a congenital condition where the opening of the penis is on the underside of the penis.

Risk Factors

- ☑ *Family history:* This condition is more common in infants with a family history of hypospadias.
- ☑ *Genetics:* Certain gene variations may play a role in disruption of the hormones that stimulate formation of the male genitals.
- ☑ *Maternal age over 35:* Some research suggests that there may be an increased risk of hypospadias in infant males born to women >35 years.
- ☑ Exposure to certain substances during pregnancy. There is some speculation about an association between hypospadias and a mother's exposure to certain hormones or certain compounds such as pesticides or industrial chemicals, but further studies are needed to confirm this.

Ideal Age for Surgery: 6–18 Months

Signs and Symptoms

- ☑ Abnormal spraying during urination
- ☑ Meatus—Urethral orifice at a location other than the tip of the penis.
- ☑ Chordae—Downward curve of the penis.
- ☑ Hooded prepuce—Only the top half of the penis is covered by foreskin.
- ☑ When the opening of the urethra is abnormal and the testicles cannot be felt on examination, the genitals may be difficult to identify as clearly male or female (*ambiguous genitalia*). In this case, further evaluation with a multidisciplinary team is recommended to determine the true anatomy and sex of the baby.

Types (Based on Meatus Location)

- ☑ Glandular, Coronal, proximal penile, distal penile, penoscrotal.

Complications

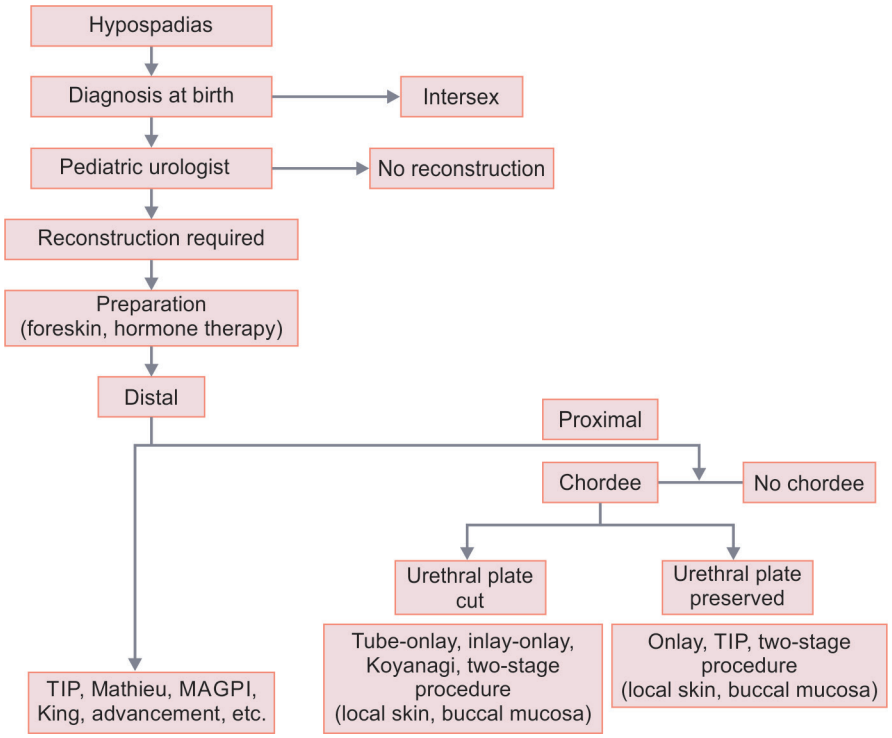
If hypospadias is not treated, it can result in:

- ☑ Abnormal appearance of the penis.
- ☑ Problems in learning to use a toilet (toilet training).
- ☑ Abnormal curvature of the penis with erection.
- ☑ Problems with impaired ejaculation.

Management

- ☑ Urethroplasty is the mainstay of treatment.
- ☑ If hypospadias is present in a new born, avoid routine circumcision.
- ☑ Correction of the chordee is an equally important aspect of hypospadias correction
- ☑ Experience of the surgeon plays an important role in the surgical outcome of this condition

Flowchart 1: Management algorithm.



(MAGPI = meatal advancement and glanduloplasty; TIP = tubularized incised plate urethroplasty).

Surgical Complications

- ☑ *Fistula:* After surgery, the child may experience a leak of urine from somewhere in the new urethra.
- ☑ Penile scaring.

Further Reading

- ☑ Belman AB. Hypospadias and chordee. In: Belman AB, King LR, Kramer SA, (Eds.). Clinical Pediatric Urology. 4th edition. London: Martin Dunitz; 2002. pp 1061-92.
- ☑ Hayashi Y, Yoshiyuki K, Mizuno K, Nakane A, Kurokawa S, Maruyama T, et al. Neo-modified Koyanagi technique for the single-stage repair of proximal hypospadias. J Ped Urol. 2007;3(3):239-42.21.
- ☑ Lam PN, Greenfield SP, Williot P. 2-stage repair in infancy for severe hypospadias with chordee: long-term results after puberty. J Urol. 2005;174(4 Pt 2):1567-72.
- ☑ Snodgrass WT, Yucel S. Tubularized incised plate for mid shaft and proximal hypospadias repair. J Urol. 2007;177(2):698-702.
- ☑ Wogelius P, Horvath-Puho E, Pedersen L, Norgaard M, Czeizel AE, Sorensen HT. Maternal use of oral contraceptives and risk of hypospadias–a population-based case-control study. Eur J Epidemiol. 2006;21(10):777-81.

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Severe Acute Malnutrition

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